15.3 Lagrange multipliers

Setup: Let $z = f(x,y)$ and $z = g(x,y)$ be functions, and k a fixed number.

Goal: Find max and min of f, <u>but</u> require them to satisfy $g(x,y) = k$.

Method of Lagrange multipliers:

(1) Find all x, y, λ such that

$$\nabla f(x,y) = \lambda \nabla g(x,y)$$

and

$$g(x,y) = k. \quad \leftarrow \text{Called the constraint.}$$

(2) Evaluate $f(x,y)$ at all points (x,y) found in (1). Biggest = max, smallest = min.

Note: The same works with $f(x,y,z)$, $g(x,y,z)$, k.

→ Do all problems.